

Conversational Agents: A Literature Review

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Abstract—Conversational agents have seen a dramatic increase in interest over the past two years. As natural language processing advances, we expect conversational agents to continue to gain interest and to inspire research. In this paper, we have reviewed 70 academic papers pertaining to conversational agents. This review categorizes these papers using various criteria and analyzes the direction of current research, then suggests several possibilities for future research.

Index Terms—conversational agents, chatbots, embodied conversational agents, natural language processing

I. INTRODUCTION

Conversational agents have been steadily gaining in popularity since their inception. A conversational agent is any computer program that generates responses in a conversation. Generally, the other participant in the conversation is a human user, although this is not always the case. In 1966, Joseph Weizenbaum created ELIZA [71], one of the first chatbots, and sparked a decades-long cultural interest in the field of conversational agents. Although ELIZA could only repeat user input in the form of a question, or occasionally give a canned response, the public's imagination became fixated on the idea and could not be shaken. Machines that could hold intelligent, human-like conversations appeared in novels, video games, and on television, and the world is increasingly convinced that science is plodding inevitably towards that future reality.

Outside the realm of science fiction, however, conversational agents have a long way to go. Although we have come a long way since the 1960s, we struggle to develop conversational agents that can consistently pass a Turing test, and agents' abilities often fall short of users' expectations [35]. Despite this, we continue to expand our understanding of human-computer interaction and improve our conversational models. In the interest of furthering this goal, we have composed this review of academic literature pertaining to conversational agents.

This paper is organized as follows: Section II describes the criteria for selecting papers for this review, Section III provides our analysis of the papers, and Section IV lays out our plans for future drafts of this review and for further research.

II. SELECTION CRITERIA

To select papers for this review, we began searching academic databases for papers using the terms “conversational agent” or “conversational agents” in their abstracts or index terms.

This search yielded 361 results from ACM. We reviewed the first 72, choosing the ones that ACM's search engine deemed “most relevant” first. We excluded two papers, one because there was no full text available, and the other because it was an analysis of science fiction. This left 70 papers that we have analyzed in the next section.

For future drafts, we hope to complete our analysis of all current ACM search results, as well as expand the search to other databases, namely ScienceDirect and IEEE, with the same search terms. It may also be worth considering adding more search terms, as the term “conversational agent” may have only recently come into use. Alternatives are discussed in Section IV.

III. ANALYSIS

For this literature review, we have reviewed 70 academic papers pertaining to conversational agents. We have categorized them by several criteria, namely the type of agent described, the application field, the design goal of the agent, the gender assigned to the agent, and the year the paper was published.

A. Type of Agent

Conversational agents can be categorized into three types, characterized by their method of communication. Although all conversational agents have common essential properties, differences in their modes of communication lead to different design and research goals.

1) *Text-based*: Text-based conversational agents, sometimes referred to as chatbots, communicate only via textual exchange. Many chatbots are implemented using existing chat applications, such as Discord and Slack. Research in text-based conversational agents focuses primarily on natural language processing and improving language models.

2) *Voice-based*: Voice-based conversational agents communicate using spoken language. Voice-based conversational agents are often designed to be virtual assistants.

3) *Embodied*: Embodied conversational agents have virtual bodies, or sometimes just faces, and are generally capable of communicating through facial expressions and gestures as well as through spoken language. Research in embodied conversational agents focuses on improving these gestures and expressions to be more human-like and appealing to the user.

Table I categorizes each paper by what type of agent it pertains to. This data is visualized in Figure 1. Over half of

the reviewed papers dealt with embodied conversational agents rather than text-based or voice-based.

Note that the a few papers were put into more than one category. [27] compares the performance of two agents, one text-based and one voice-based. [55] compares a text-based agent and an embodied agent. [22] compares a voice-based agent and an embodied agent. Finally, [31] and [2] each describe a voice-based agent that also has a text-only interface.

TABLE I
PAPERS BY AGENT TYPE

Text-based	[68] [54] [3] [17] [53] [69] [27] [65] [72] [18] [25] [58] [47] [55] [28] [62] [40] [31] [2] [32] [33] [37]
Voice-based	[35] [27] [59] [51] [52] [22] [31] [2] [49] [12] [61]
Embodied	[41] [67] [19] [16] [21] [44] [24] [42] [26] [20] [36] [63] [57] [70] [66] [48] [73] [39] [56] [50] [11] [4] [55] [38] [1] [13] [46] [22] [14] [64] [23] [10] [45] [5] [8] [6] [43] [7] [60]
Unspecified	[34] [15] [30]

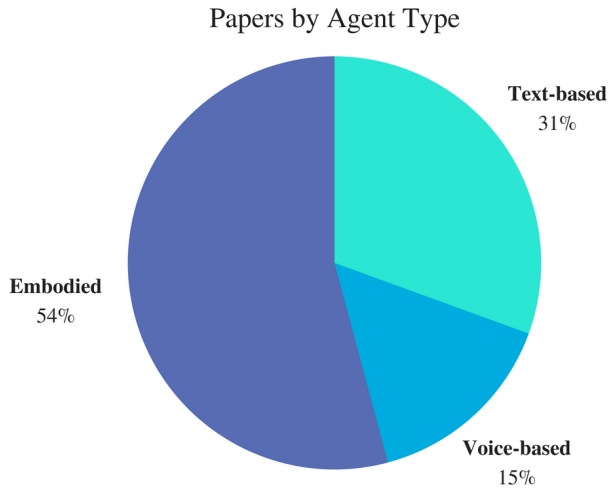


Fig. 1. Papers categorized by type of conversational agent.

B. Application

We are interested in recognizing which fields are most frequently addressed in writings about conversational agents. Most papers, however, do not go into detail about which fields would most benefit from their research. The few that outline possibilities seem to focus on either educational use or financial application. Nonetheless, the vast majority of papers specify no application field. We speculate that this is due to the nature of conversational agents. Most are deliberately designed to be flexible, and a single agent can often be adapted for use in several different fields.

C. Design Goal

Research on conversational agents seems to be segregated into two areas of interest, characterized by the perceived purpose of the conversational agent. A conversational agent,

in general, exists either to assist the user with some task, or to build rapport with the user. Depending on the researchers' goals and expectations, their design may focus on one or the other, although there is often overlap between the two approaches.

When designing conversational agents to assist the user with a task, researchers generally focus on the agent's ability to interpret instructions correctly, to respond appropriately in the context for which the agent is designed, and in some cases to incorporate specific knowledge bases into its conversation. When researchers design with the goal of building rapport with the user, on the other hand, they tend to focus on giving the agent a semblance of personality, improving gestures and expressions to be more human-like, and increasing the scope of the agent's conversational abilities.

For the purposes of this review, we have categorized each paper by its focus on either assisting the user or building rapport with the user. Although many papers address aspects of both concerns, we have assigned each paper to a single category in the interest of simplicity. This categorization is based on our reading of the papers and may be somewhat subjective for individual papers, but is meant to indicate overall trends in research.

Table II shows the categorization of each paper by its design goals. This data is visualized in Figure 2. Papers were about evenly split between the two approaches. This indicates that there is significant interest in conversational agents that both assist the user with a task and can build rapport with the user.

TABLE II
PAPERS BY DESIGN GOAL

Assist	[68] [3] [17] [42] [53] [69] [35] [59] [65] [72] [18] [39] [34] [58] [51] [28] [52] [62] [13] [22] [31] [14] [2] [32] [64] [12] [10] [8] [33] [61] [7]
Build rapport	[41] [54] [67] [19] [16] [21] [44] [24] [42] [26] [20] [36] [63] [57] [70] [66] [27] [48] [73] [56] [50] [25] [15] [11] [4] [47] [55] [38] [1] [40] [46] [49] [23] [45] [5] [30] [6] [37] [43] [60]

D. Gender

We are interested in discovering how perceptions of the conversational agent's gender affects research, both in terms of users' interactions with agents as well as researchers' interpretations of the results. Unfortunately, most researchers do not take gender perceptions into account when reporting their findings, despite evidence that perceived gender has an effect on interaction [9], [29]. Since most papers make no mention of the designated gender of the conversational agent, we cannot hope to analyze overall trends. We hope to see future research address this issue.

E. Publication Year

Currently, the 70 papers we have analyzed range in publication date from 2000 to 2018. Figure 3 shows the distribution of the current selection of papers over publication year. Prior

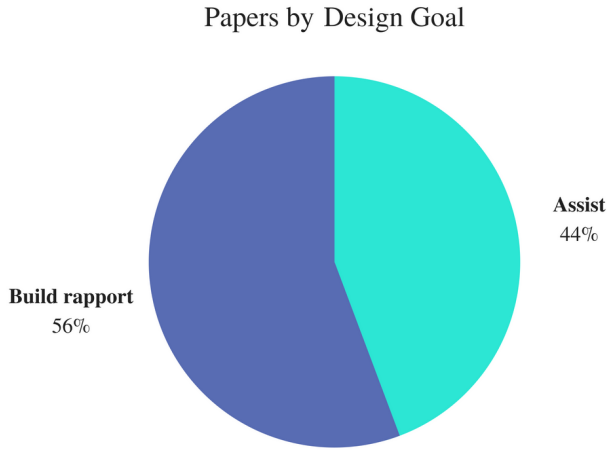


Fig. 2. Papers categorized by design goals.

to 2017, no more than 5 papers were published per year pertaining to conversational agents. In the past two years, there has been a huge increase in interest, jumping to 11 papers published in 2017 and 13 so far in 2018. This indicates that conversational agents have very quickly gained interest in academic circles over the past two years.

Figure 4 shows the distribution of papers over publication year, separated by the type of agent described as defined in Section III-A. This indicates that although all conversational agents have seen an increase in interest over the last two years, the effect has been the most dramatic in text-based conversational agents.

Figure 5 shows the distribution of papers over publication year, separated by the design goals of the agent, as defined in Section III-C. Both approaches to designing conversational agents have seen an increase in interest in recent years, but papers on assisting agents have increased dramatically starting in 2017, while the increase in papers on rapport-building agents has been more gradual.

It is unclear whether date of publication is a factor in the search engine's criteria for "relevance." It is possible that more recent papers were presented first and are therefore more numerous in the first 70 results than in the rest of the search results. To confirm our findings, we will need to complete our analysis of all 361 papers found by the search terms.

IV. FUTURE WORK

For future drafts of this review, we would like to expand our analysis to include all the search results from ACM, as well as expand the search to ScienceDirect and IEEE databases. We may also need to consider expanding the search terms, as we want our review to encompass a broad history of conversational agents, but the term "conversational agent" may be relatively a relatively new one. Weizenbaum's 1966 paper on ELIZA [71], which we have cited as one of the first conversational agents, does not actually use the term, calling it instead a program capable of "natural language conversation."

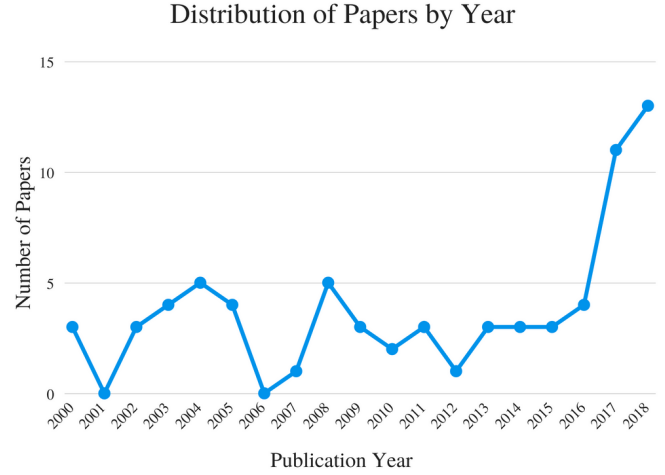


Fig. 3. Papers on conversational agents distributed by publication year.

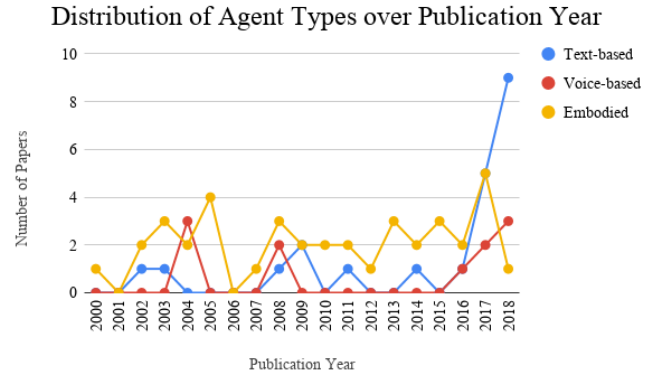


Fig. 4. Types of conversational agents distributed by publication year.

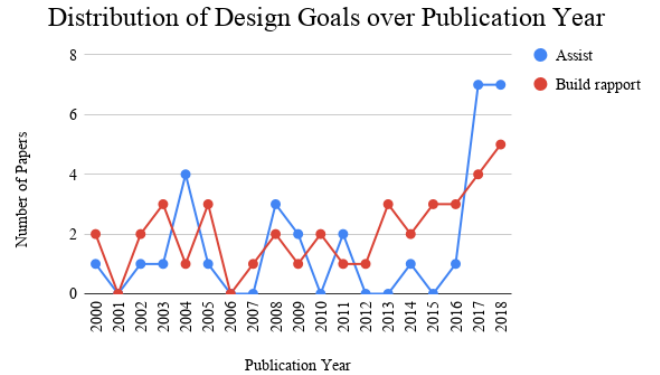


Fig. 5. Design goals of conversational agents distributed by publication year.

We may consider expanding our search to include this term, and may need to research what other terms may have been used to describe the same concept.

We have discussed several areas in the current research on conversational agents that have seen an increase in recent years. In particular, there seems to have been an increase in interest in text-based conversational agents and in agents that are designed to assist the user with some task. It may be advantageous for future research to take advantage of this increase in interest to explore areas of conversational agents that have not seen as much attention.

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